

Module 5

Mapping Distance and Density

Distance and Density Analysis using ArcGIS Spatial Analyst

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Registration No: **405125**

Lab No: **Lab 3**

Software: **ArcMap / ArcGIS Spatial Analyst**

1. Objectives

The purpose of this laboratory exercise was to gain practical experience with the distance and density functions available in the ArcGIS Spatial Analyst extension. On completing the exercise a student should be able to:

- Create straight-line distance, direction, and allocation surfaces.
- Create cost-weighted distance, direction, and allocation surfaces.
- Perform a least-cost path analysis between a source and a destination.
- Create density surfaces using the simple and the kernel methods.
- Estimate density using the attribute values of point features.

2. Introduction

Spatial Analyst distance functions allow an analyst to determine the nearest location of a feature or the least-cost path to a destination, while density functions reveal where features are most and least concentrated. Two conceptually different distance surfaces are used in this module: one that measures distance "as the crow flies" (straight-line distance) and one that measures distance in terms of the cost of travel across a surface (cost-weighted distance).

Every distance surface is calculated from a source, which is the location of some object of interest such as a hospital or a power plant. Where more than one source exists, the calculation is based on the nearest one. Density surfaces, by contrast, store the number of features per unit of area in each cell, making them ideal for visualising spatial patterns such as crime hotspots that might otherwise remain hidden in point data.

This report documents the four tasks completed during the exercise. Each task is presented with a short description of the method used, followed by the ModelBuilder diagrams and map outputs produced in ArcMap.

3. Software and Data

All work was carried out in ArcMap with the Spatial Analyst extension enabled, using the Distance.mxd and Density.mxd map documents supplied with the module. Workflows were assembled in ModelBuilder so that each analysis could be run, reviewed and, where necessary, re-executed as a repeatable model. Custom tools were organised in a dedicated DistanceTools toolbox.

4. Methodology and Results

4.1 Task 1 Straight-Line (Euclidean) Distance

The first task applied the Straight Line Distance and Straight Line Allocation functions to an air-ambulance scenario, in which the objective was to describe how far every location in the study area is from its nearest hospital. A straight-line distance surface measures the shortest distance from the centre of each cell to the centre of the nearest source cell, in the same way a ruler measures the distance between two points. The allocation function then assigns every cell to its nearest hospital, partitioning the study area into service zones.

The hospitals were set as the source and the Euclidean Allocation tool was run in ModelBuilder to generate allocation, distance, direction and back-direction rasters. The raw distance surface was converted into miles

with the Divide tool and then grouped into readable classes with the Reclassify tool, producing a clear map of distance-to-nearest-hospital in miles.

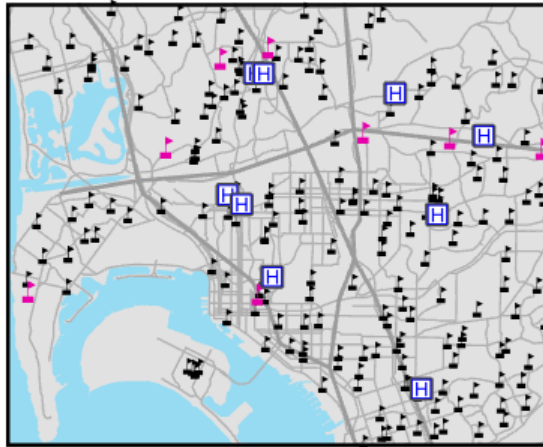


Figure 1. Study-area map in Distance.mxd showing hospital locations and points of interest.

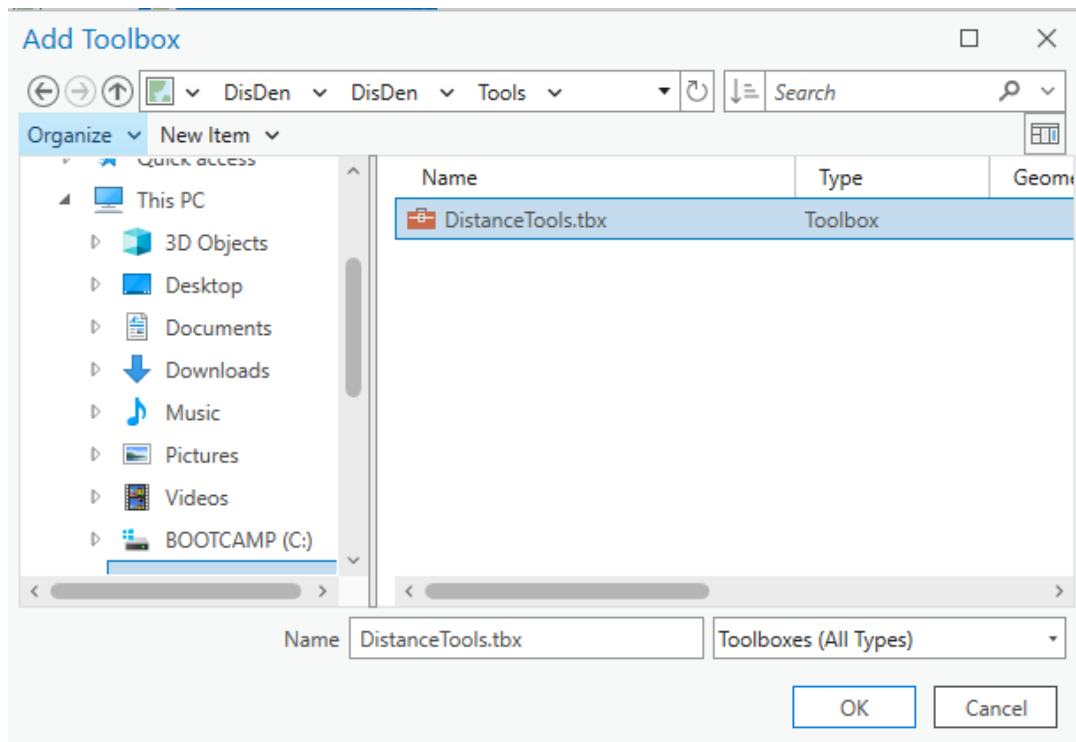


Figure 2. Adding the custom Distance Tools toolbox (DistanceTools.tbx) to ArcToolbox.

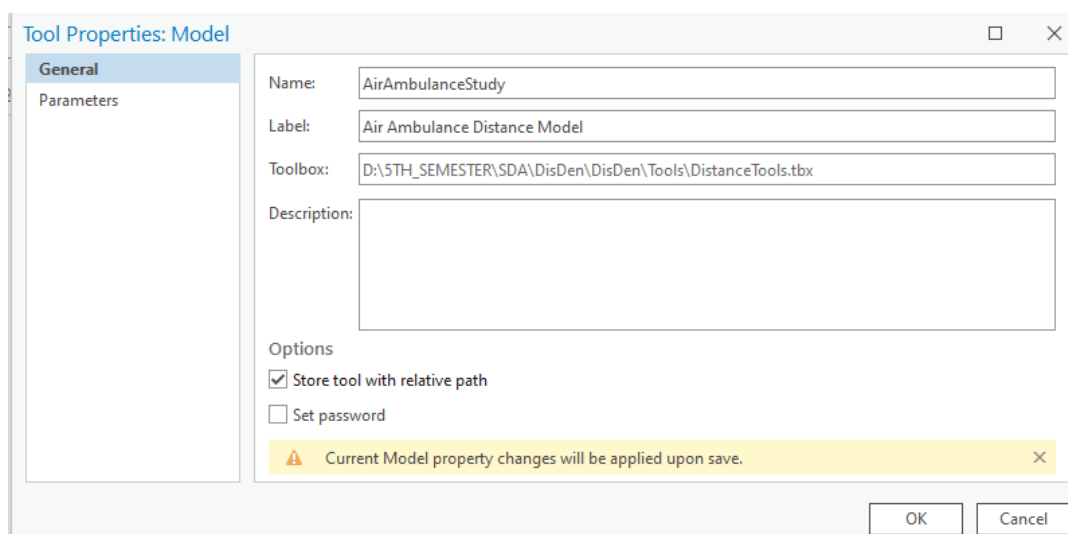


Figure 3. Creating the Air Ambulance Study model in ModelBuilder.

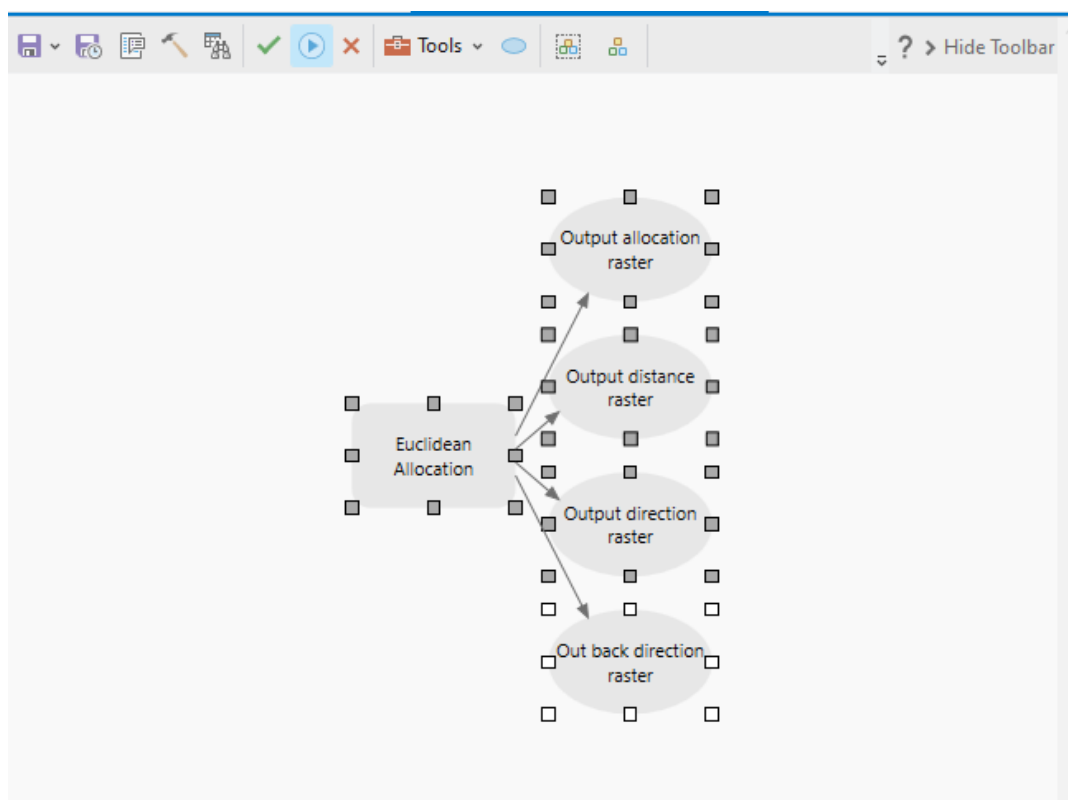


Figure 4. Euclidean Allocation tool and its four output rasters: allocation, distance, direction and back-direction.

Euclidean Allocation [X]

This tool is **deprecated** and will be removed in a future release.
 The **Distance Allocation** tool provides enhanced functionality or performance. [X]

Parameters | **Environments** | **Properties** [?]

Input raster or feature source data
 Hospital [Folder] [Pencil] [v]

Source field
 ID [v] [Gear]

Output allocation raster
 euclideanforout [Folder]

Input raster or feature barrier data
 [Folder] [Pencil] [v]

Input value raster
 [Folder] [v]

Maximum distance
 [Text Box]

Output cell size
 100 [Folder] [v]

Distance Method
 Planar [v]

Output distance raster
 outputdistance [Folder]

Output direction raster
 outputdirection [Folder]

Out back direction raster
 outbackrasteroutput [Folder]

[OK]

Figure 5. *Euclidean Allocation tool parameter dialog (hospital set as the source).*

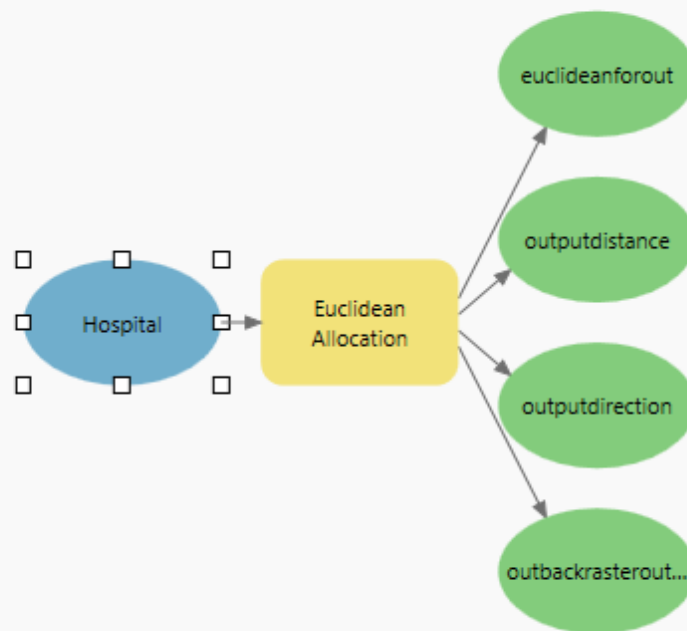


Figure 6. Model diagram: the Hospital source feeding the Euclidean Allocation tool.

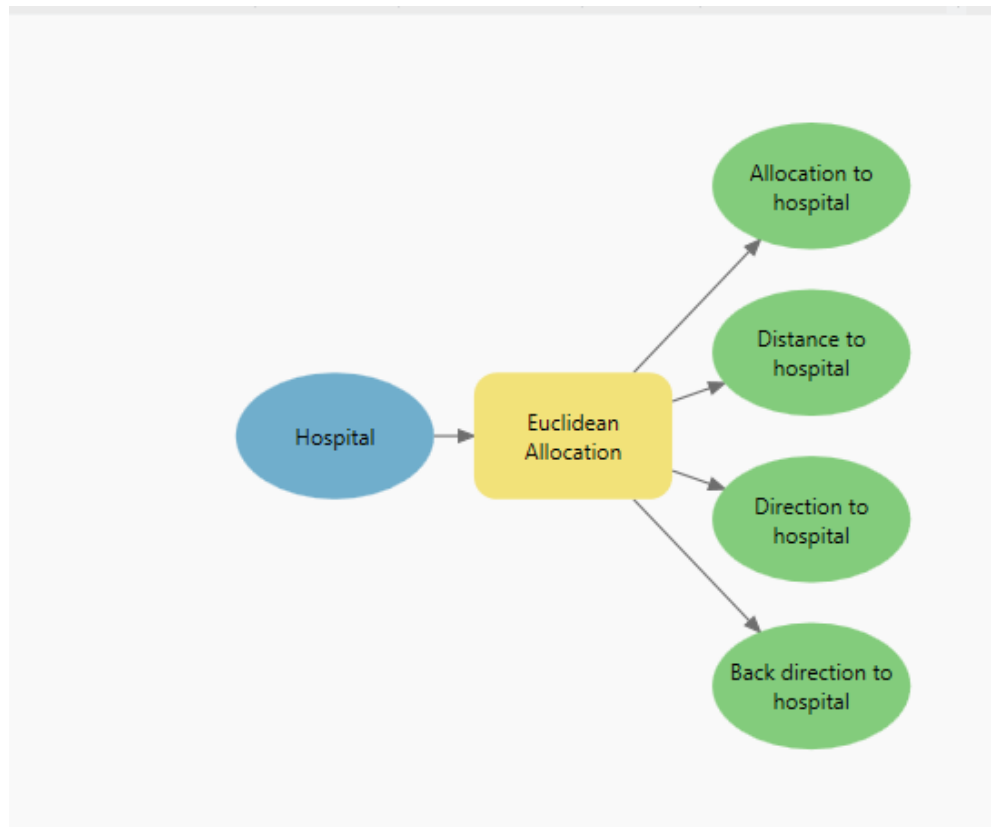


Figure 7. Model outputs: allocation, distance, direction and back-direction to the nearest hospital.

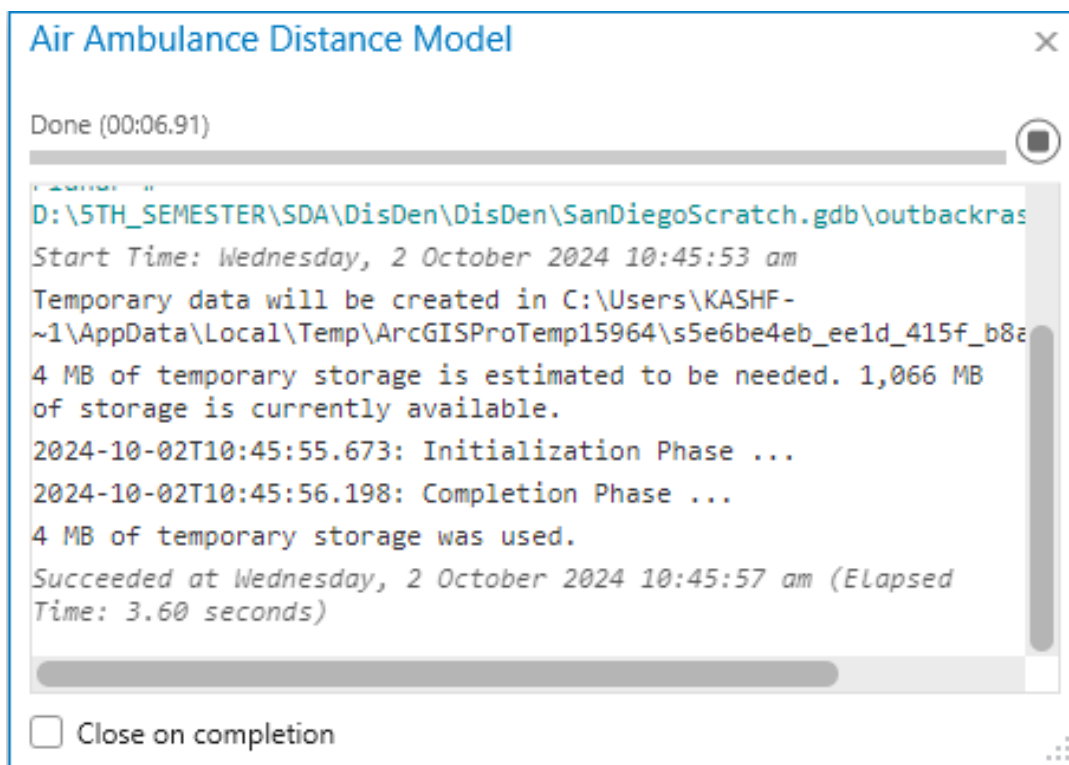


Figure 8. Air Ambulance Distance Model run log confirming successful execution.

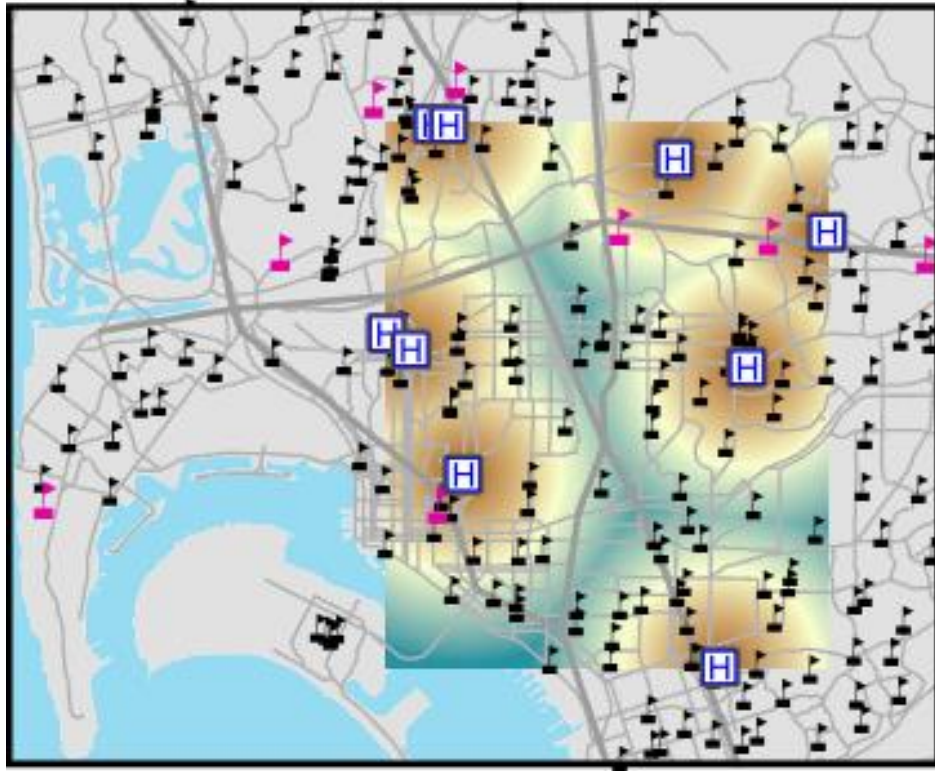


Figure 9. Straight-line allocation surface: the study area partitioned into service zones around each hospital.

Study Area - 0	
OBJECTID	1
Id	0
Shape_Length	256619.743983
Shape_Area	4073027638.016237

Figure 10. Study Area attribute table; Shape_Area is used to help convert distance units.

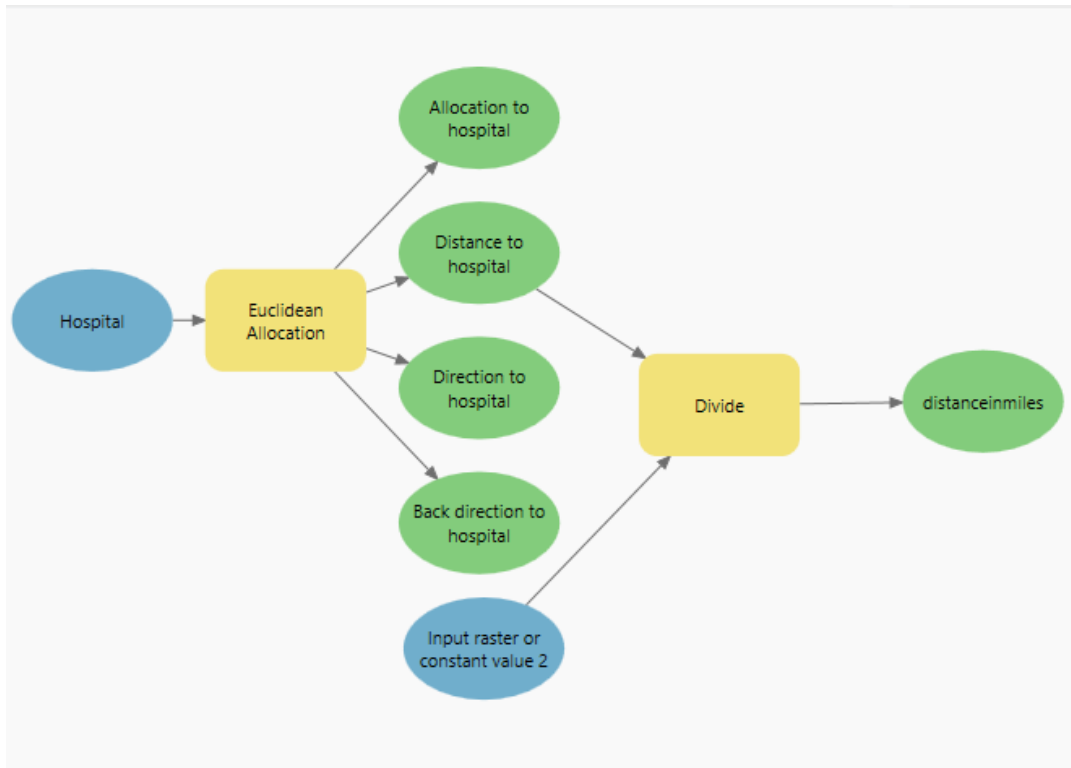


Figure 11. Model extended with a Divide tool to convert the distance surface into miles.

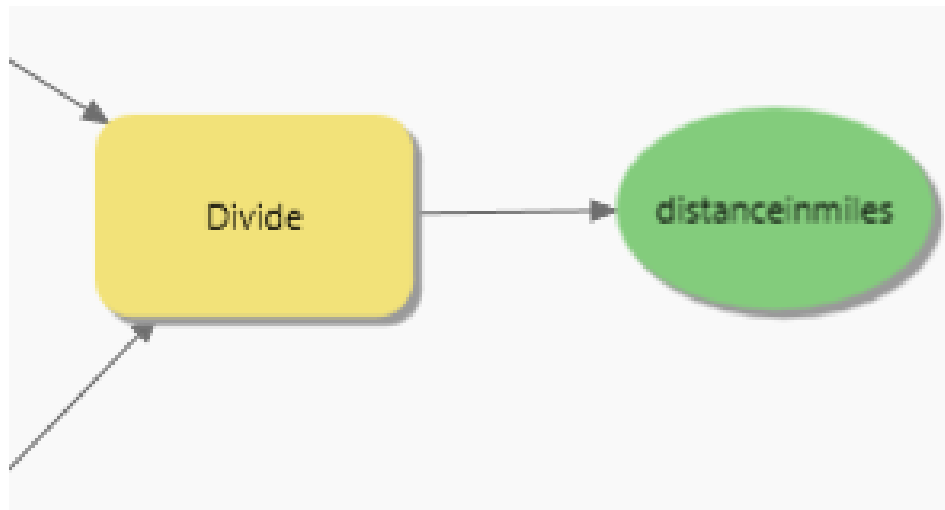


Figure 12. Divide tool producing the distance-in-miles raster.

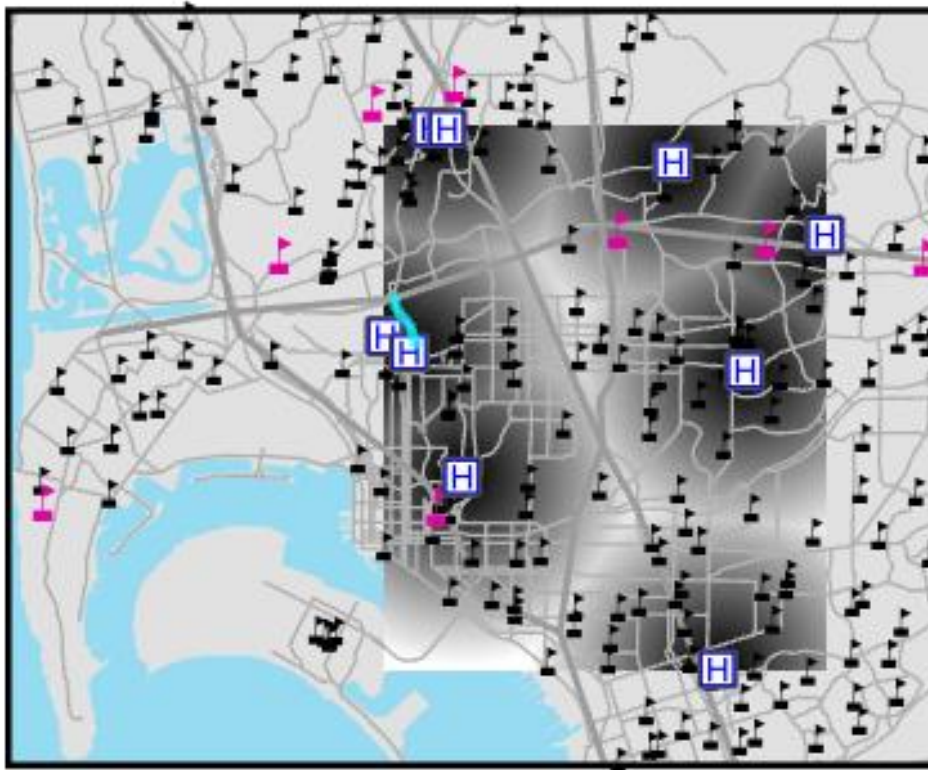


Figure 13. Straight-line (Euclidean) distance surface in greyscale: distance from every cell to the nearest hospital.

Reclassify
×

Parameters
Environments
Properties
?

Input raster
distanceinmiles
📁

Reclass field
VALUE
⚙️

* Reclassification

Reverse New Values

Value	New

Classify
Unique
📁
💾
🗑️

Output raster
Reclass_dist1
📁

☐ Change missing values to NoData

OK

Figure 14. Reclassify tool used to group the distance-in-miles values into classes.

Classify [X] []

Field: VALUE

Method: Defined Interval ▾

Interval size: 1

Classes: 3 ▾

Classes | Histogram

More ▾

Upper value	
≤ 1.0	
≤ 2.0	
≤ 3.0	

OK Cancel

Figure 15. Classify dialog using a defined-interval classification scheme.

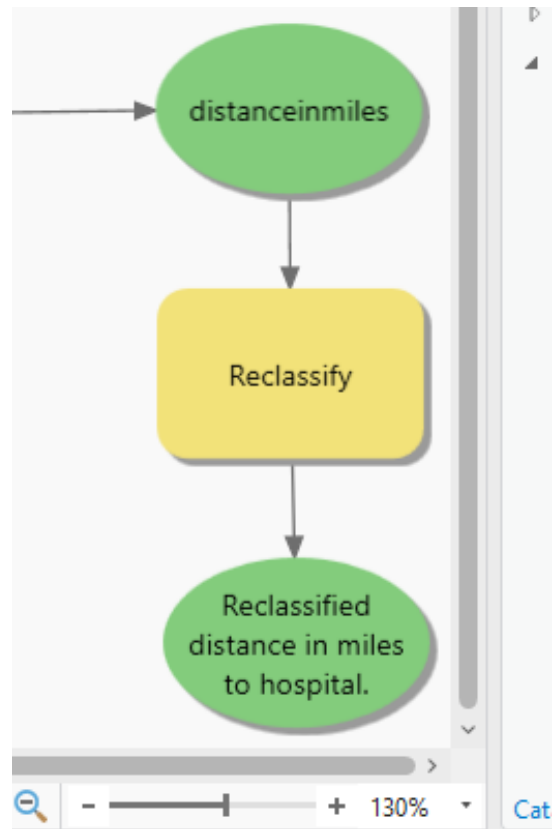


Figure 16. Model: distance-in-miles reclassified into distance-to-hospital classes.

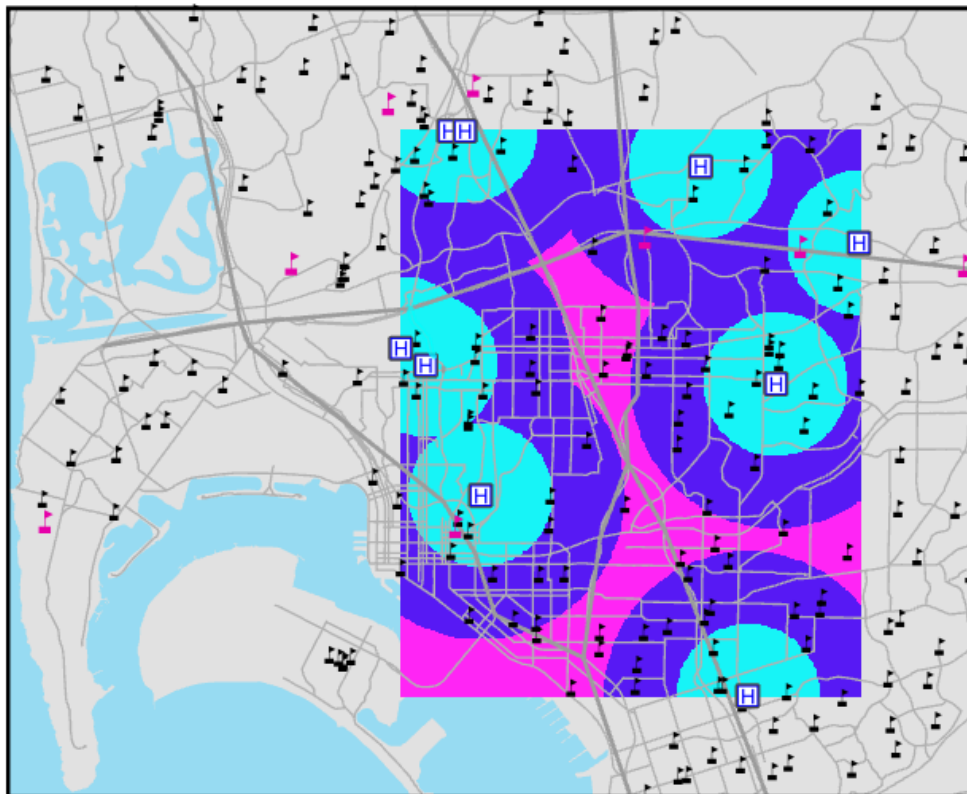


Figure 17. *Reclassified straight-line distance surface (miles to the nearest hospital).*

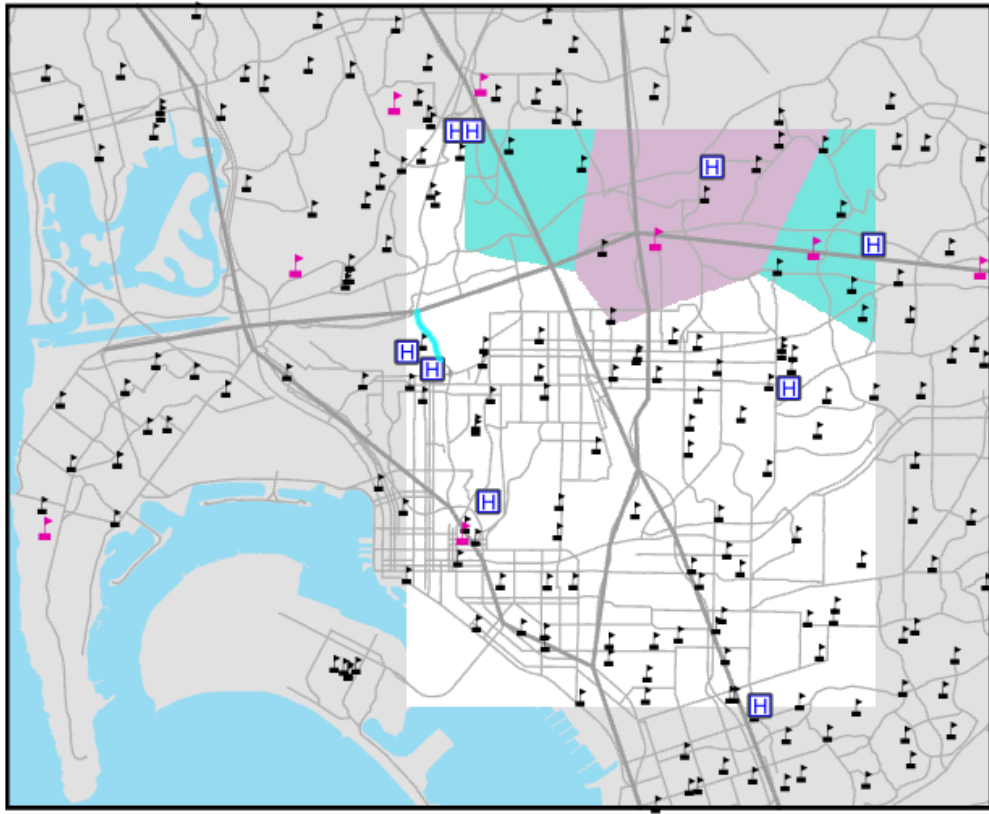


Figure 18. *Straight-line allocation surface showing hospital service areas.*

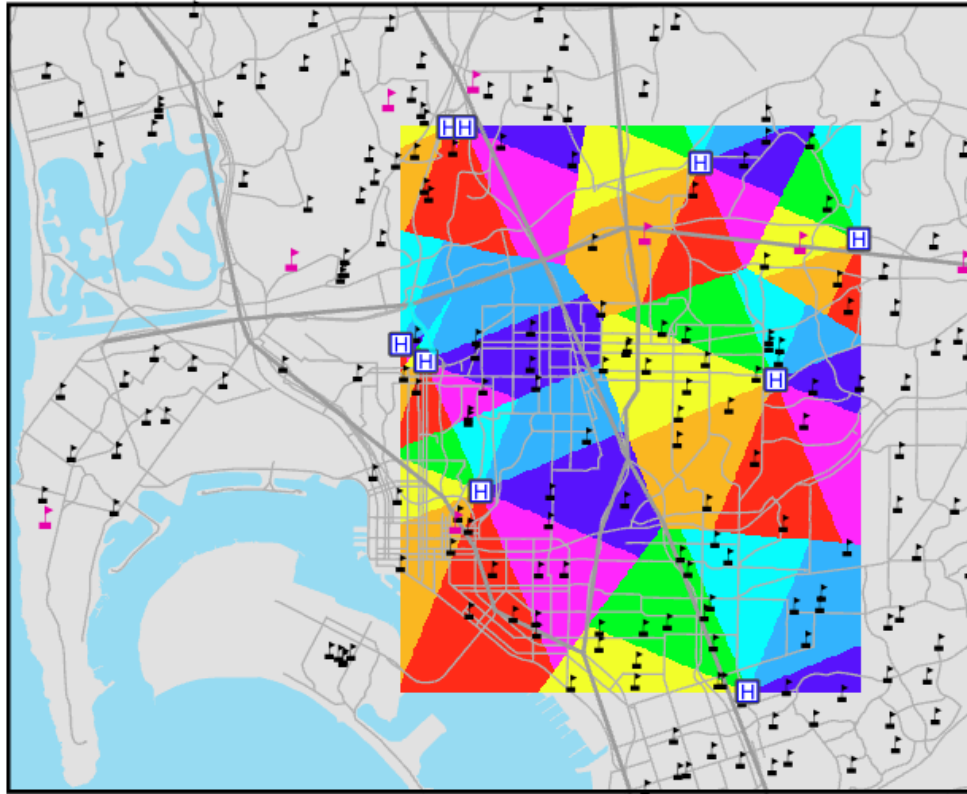


Figure 19. Combined distance and direction surface visualisation for the air-ambulance scenario.

4.2 Task 2 — Cost-Weighted Distance and Least-Cost Path

The second task moved from straight-line distance to cost-weighted distance in order to route a new power line from the Otay Valley Power Plant to the Jamul Substation. Unlike a straight-line surface, the cell values in a cost-weighted surface are not distances but accumulated costs of travel: the function evaluates the neighbours of each cell, multiplies the average cost between a pair of cells by the distance between them, and moves outward from the source always choosing the lowest-cost cell.

A single cost surface was required as input, so two factors were combined. The terrain slope was derived from the DEM and reclassified so that steeper ground carried a higher cost, and the land-use layer was reclassified so that difficult or protected land types carried a higher cost. The two ranked rasters were added with the Plus tool to create the final COST surface. The Cost Distance tool then produced cost-distance and cost-direction surfaces from the power plant, and finally the Cost Path tool traced the least-cost path — the path of least resistance — from the substation back to the plant.



Figure 20. *Digital elevation model (DEM) of the study area with the proposed power-line corridor.*



Figure 21. *Terrain view showing the source point (yellow) and the routing corridor.*



Figure 22. *Close-up of the terrain along the proposed power-line route.*



Figure 23. *Terrain with source (yellow) and destination (red) markers for the routing analysis.*

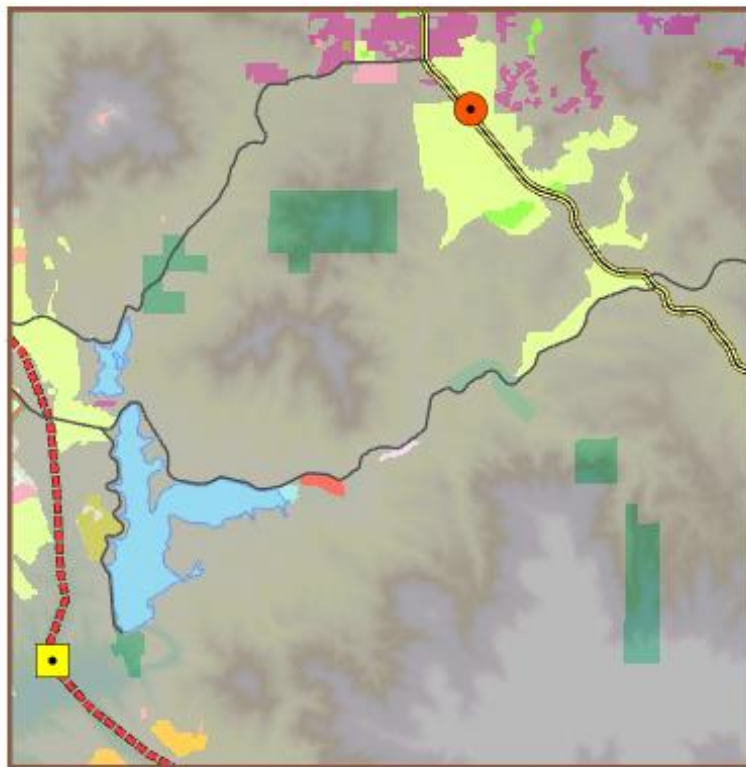


Figure 24. Land cover and terrain along the corridor between the two facilities.

Tool Properties: Model

General

Parameters

Name: Powerline

Label: Powerline Least Cost Path

Toolbox: D:\5TH_SEMESTER\SDA\DisDen\DisDen\Tools\DistanceTools.tbx

Description: This model finds the least-cost path for a proposed power transmission line between a fictional power plant site (Otay Valley Power Plant) and substation (Jamul Substation) in Southern California

Options

☒ Store tool with relative path

☐ Set password

Current Model property changes will be applied upon save.

OK Cancel

Figure 25. Powerline Least Cost Path model properties in ModelBuilder.

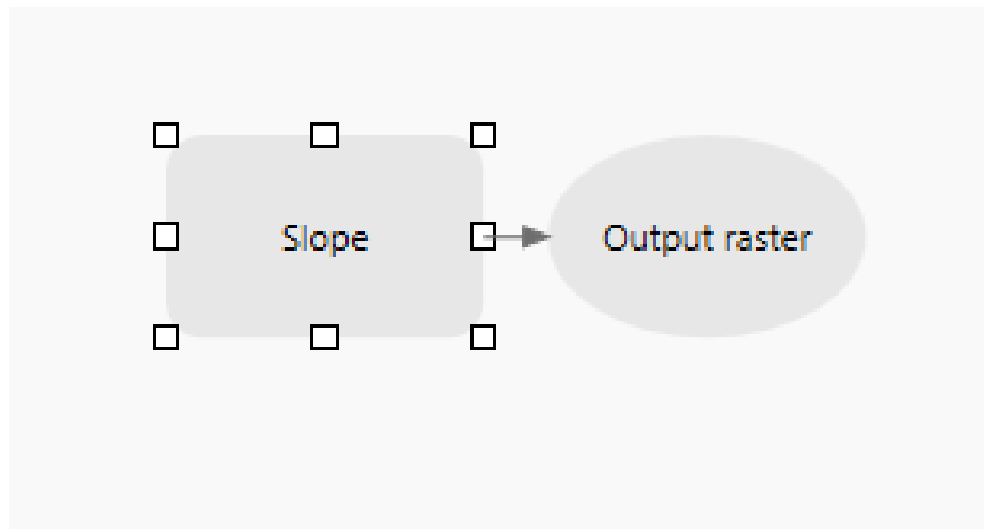


Figure 26. Initial model: Slope tool feeding an output raster.

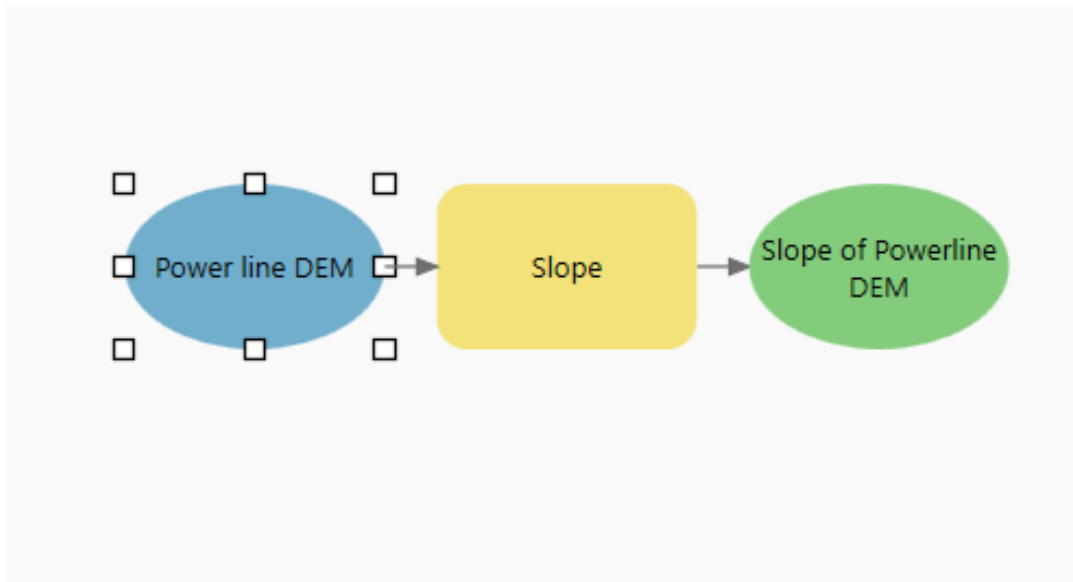


Figure 27. Model: Power line DEM to Slope to Slope of Powerline DEM.

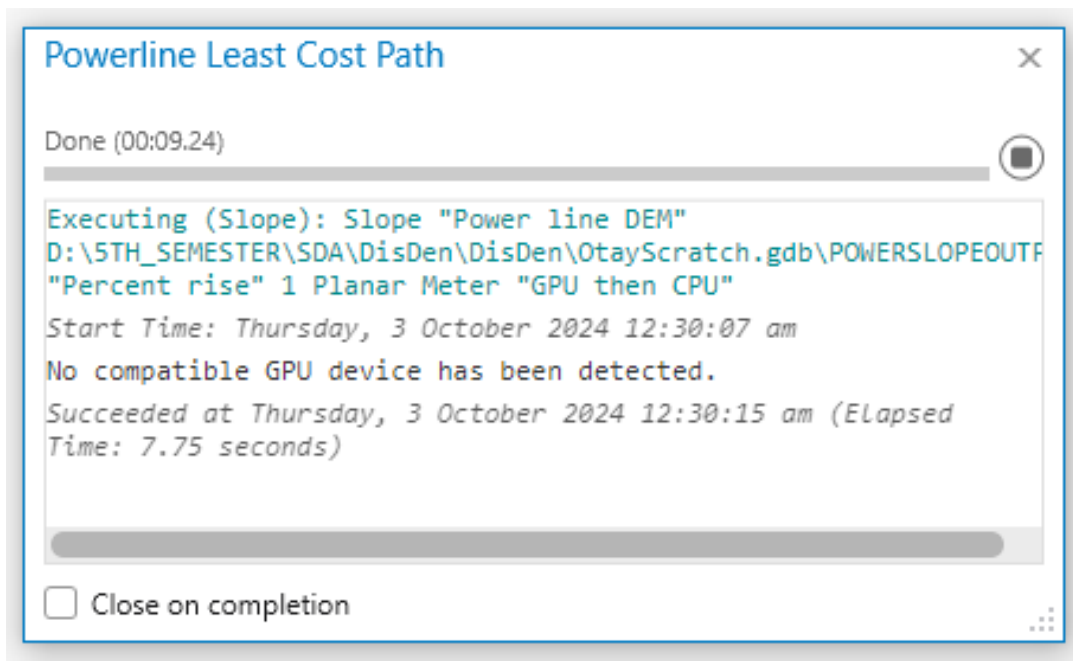


Figure 28. Slope tool run log (executed successfully).

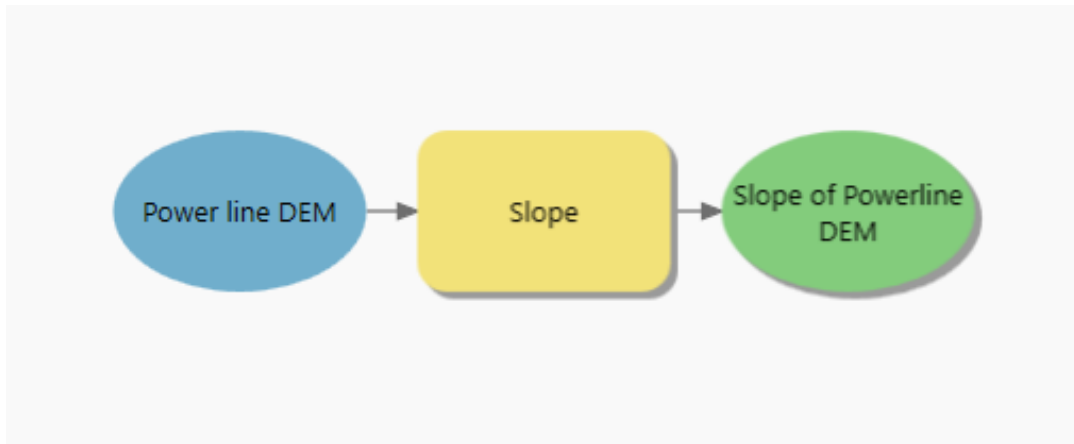


Figure 29. Completed Slope model producing the slope-of-terrain raster.

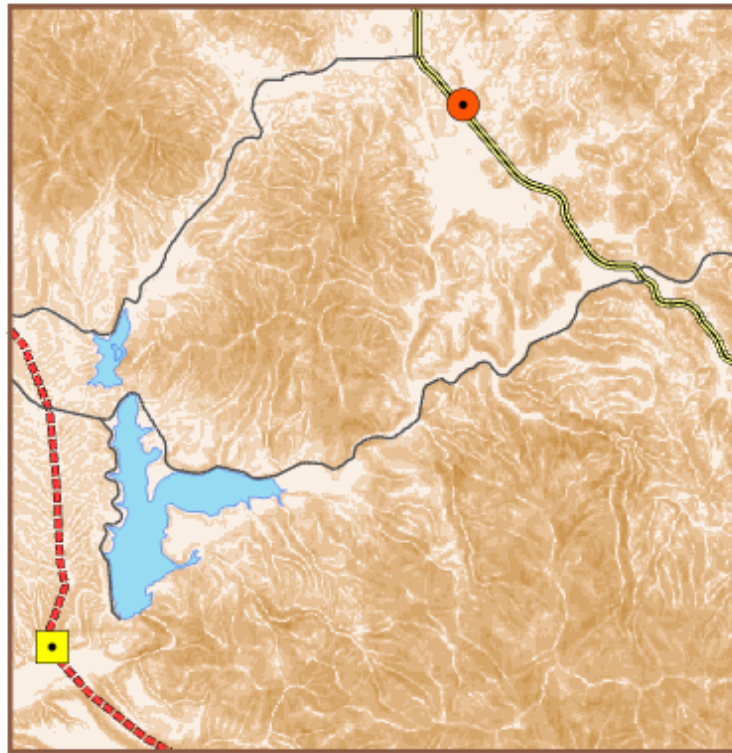


Figure 30. Slope surface derived from the terrain DEM.

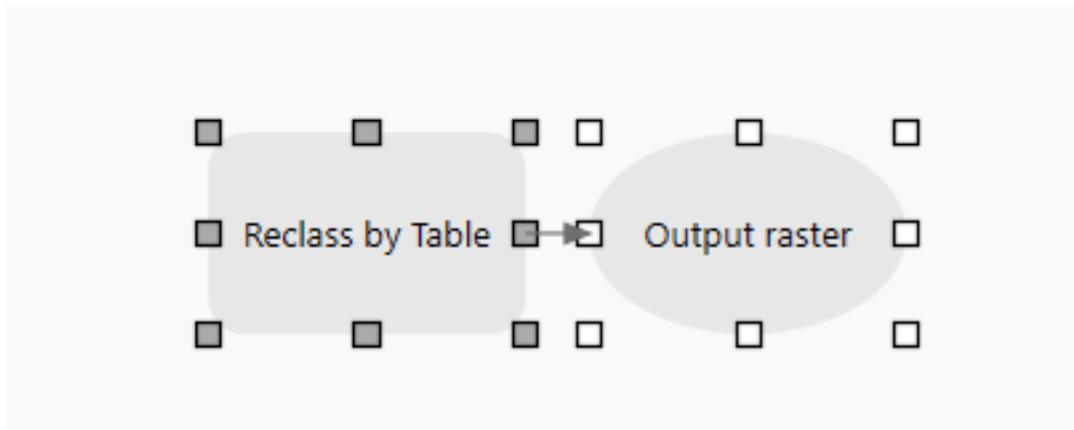


Figure 31. *Reclass by Table tool added to rank slope values by travel cost.*

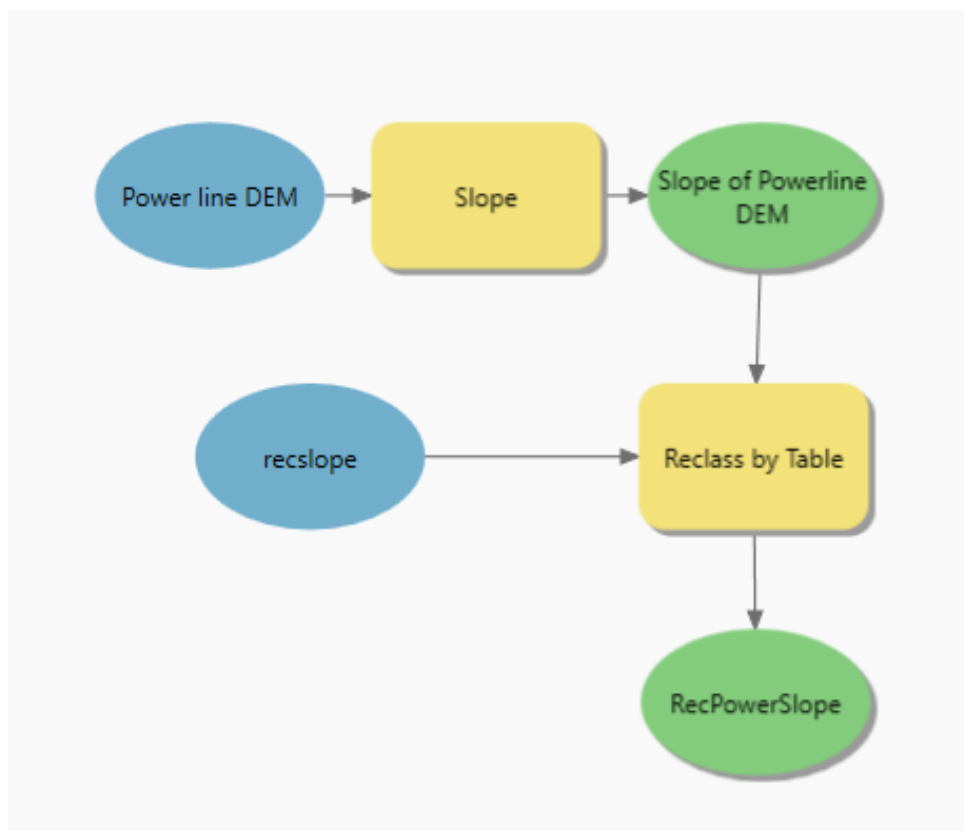


Figure 32. *Model: slope reclassified into RecPowerSlope (ranked slope cost).*

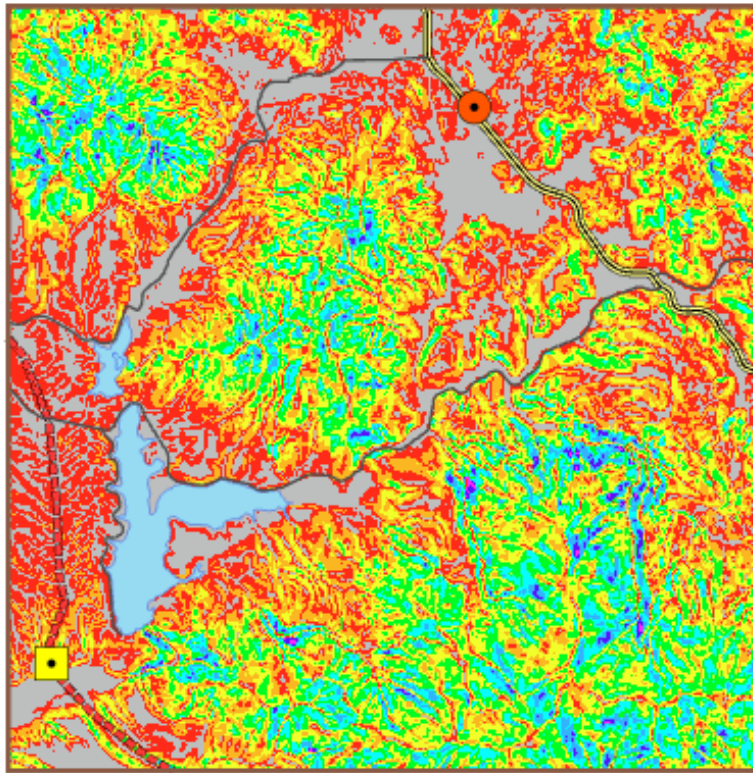


Figure 33. *Reclassified slope-cost surface.*

Symbology - RecPowerSlope:Rec... ▾ 🔍 ✕



Primary symbology

Unique Values ▾

Field 1 Value ▾

Color scheme  ▾

Values Mask

    More ▾

Color	Value	Label
	1	1
	2	2
	3	3
	4	4
	5	5

Activate Windows
Go to Settings to activate Windows.

Figure 34. *Symbology settings applied to the RecPowerSlope cost surface.*

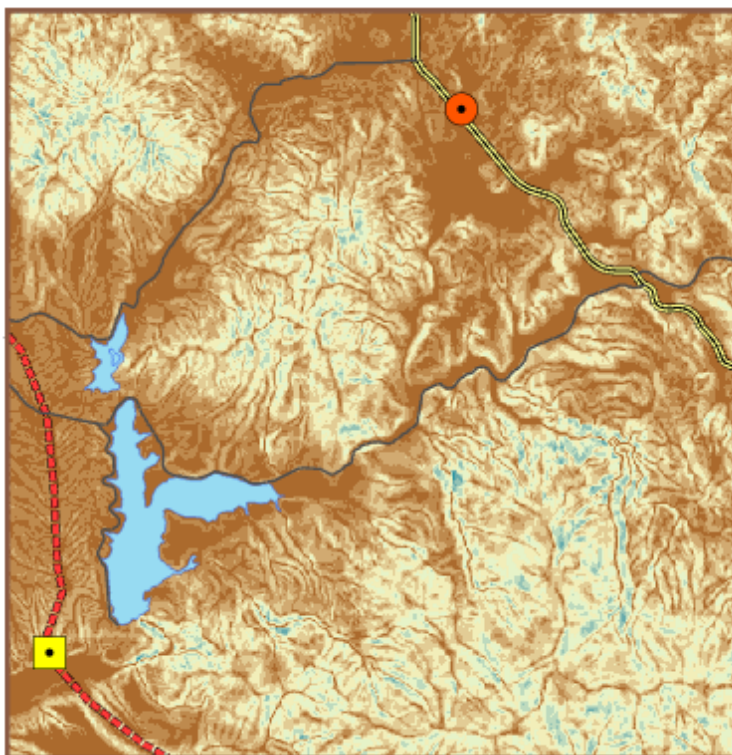


Figure 35. *Reclassified slope-cost surface displayed over the terrain.*



Figure 36. Land-use layer legend (Lutype categories) used to assign travel cost.

Reclassify [X]

Parameters | Environments | Properties [?]

Input raster
Land use [v] [Folder icon]

Reclass field
Lutype [v] [Gear icon]

Reclassification

Reverse New Values

Value	New
Airstrips	1
Communications	2
Field crops	3
Fire/Police stations	4
Golf courses	5
Heavy industrial	6
Intensive agriculture	7
Jails/Prisons	8
Lakes/Reservoirs	9
Landscape open space	10
Marinas	11
Multi-family residential	12
Office - low rise	13
Olympic training center	14
Open space preserves	15
Orchards/Vineyards	16
Post offices	17
Recreation	18
Retail/Strip commercial	19
Rural residential	20
Single family residential	21
Vacant, graded	22

Classify Unique [Folder icon] [Save icon] [Delete icon]

Output raster
RecLandUse [Folder icon]

☐ Change missing values to NoData

OK

Figure 37. *Reclassify tool ranking each land-use type by relative cost.*

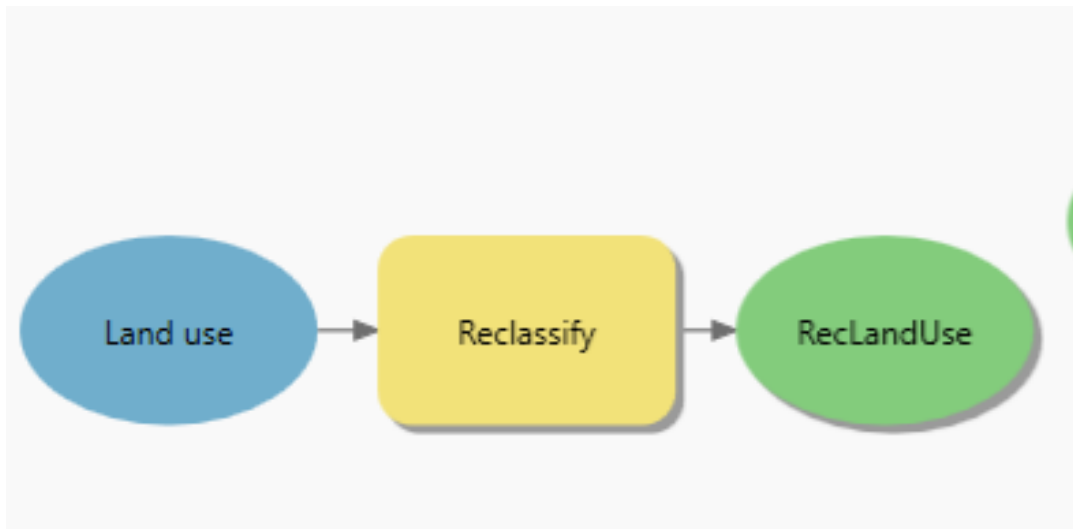


Figure 38. *Model: Land use to Reclassify to RecLandUse.*

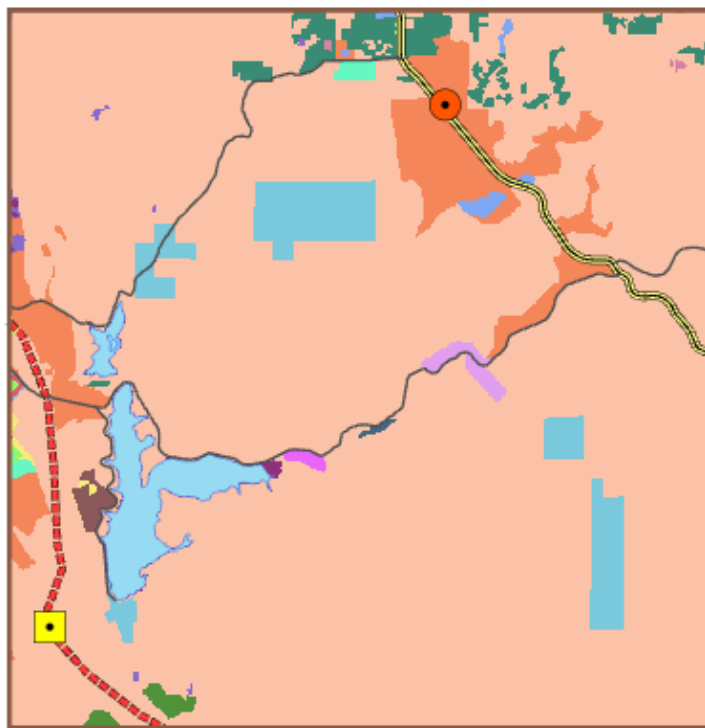


Figure 39. *Reclassified land-use cost surface.*

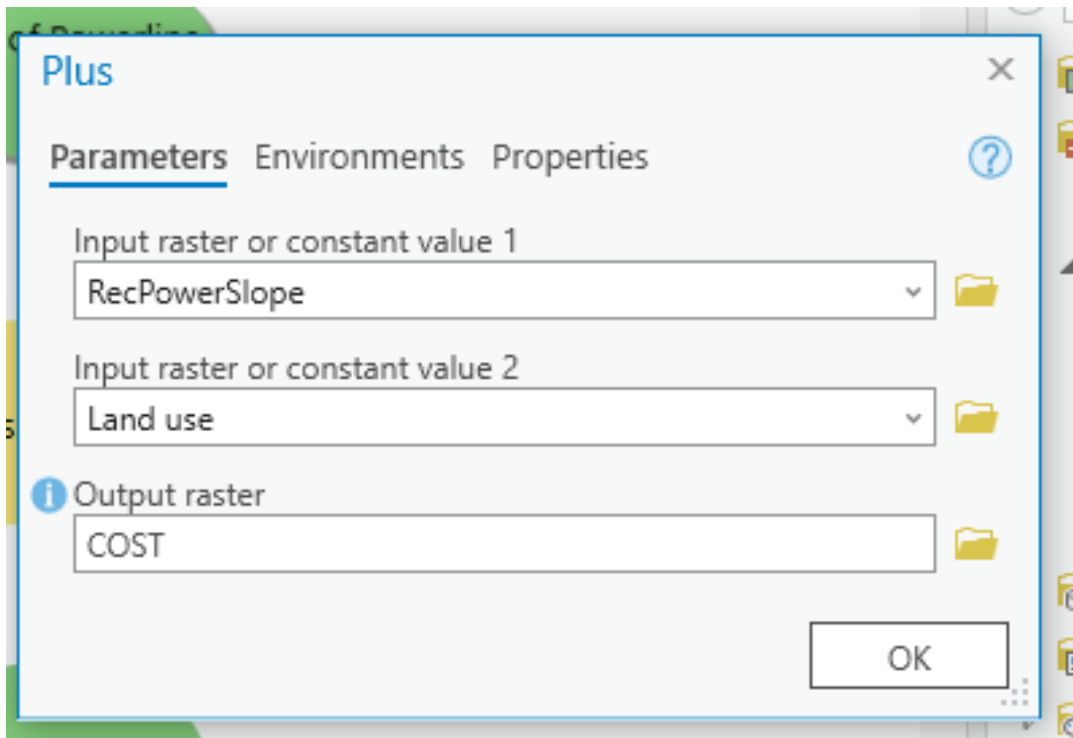


Figure 40. Plus tool combining the slope-cost and land-use-cost rasters.

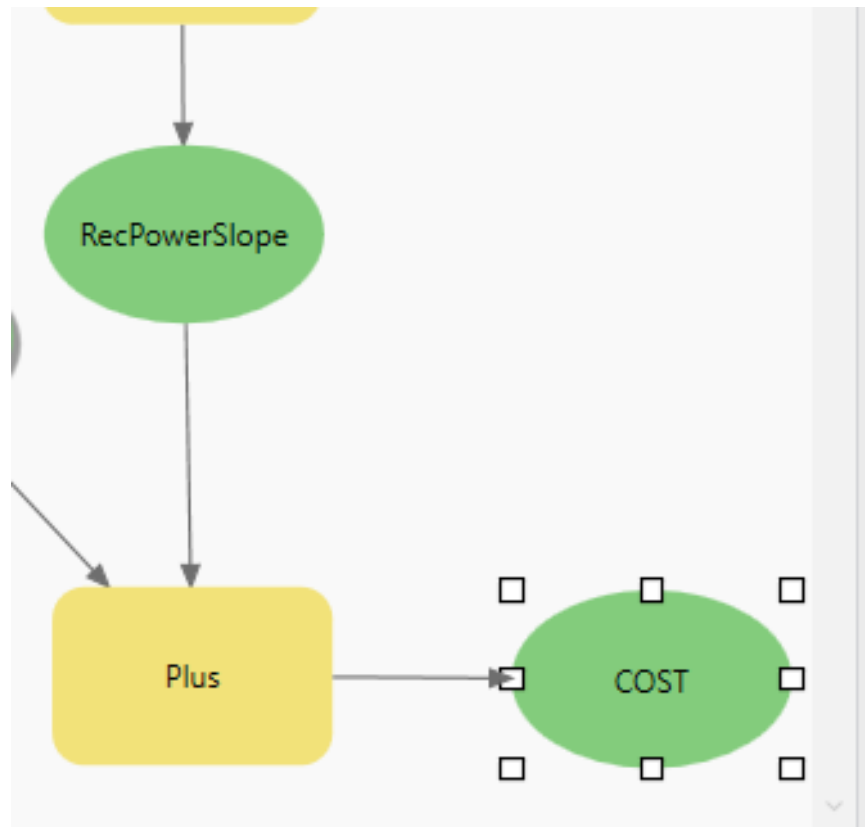


Figure 41. Model: RecPowerSlope plus Land use, via the Plus tool, producing COST.

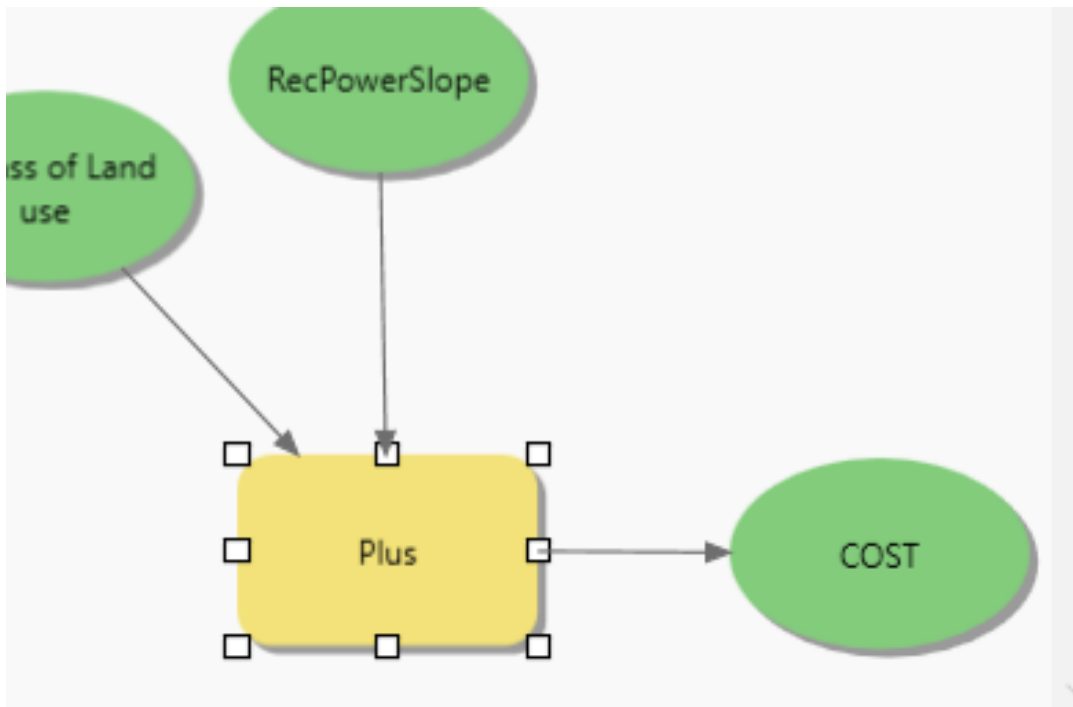


Figure 42. Full cost model producing the final combined COST surface.

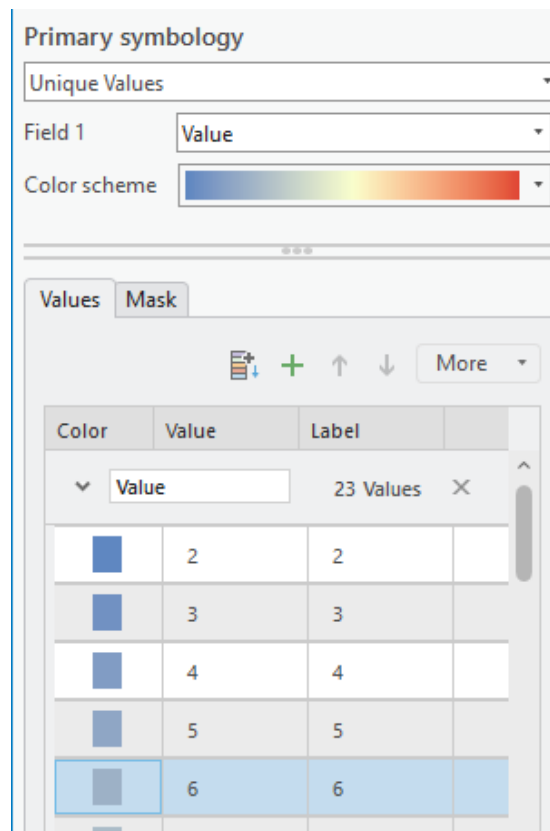


Figure 43. Symbology of the final cost-surface values.



Figure 44. *Final combined cost surface for the study area.*

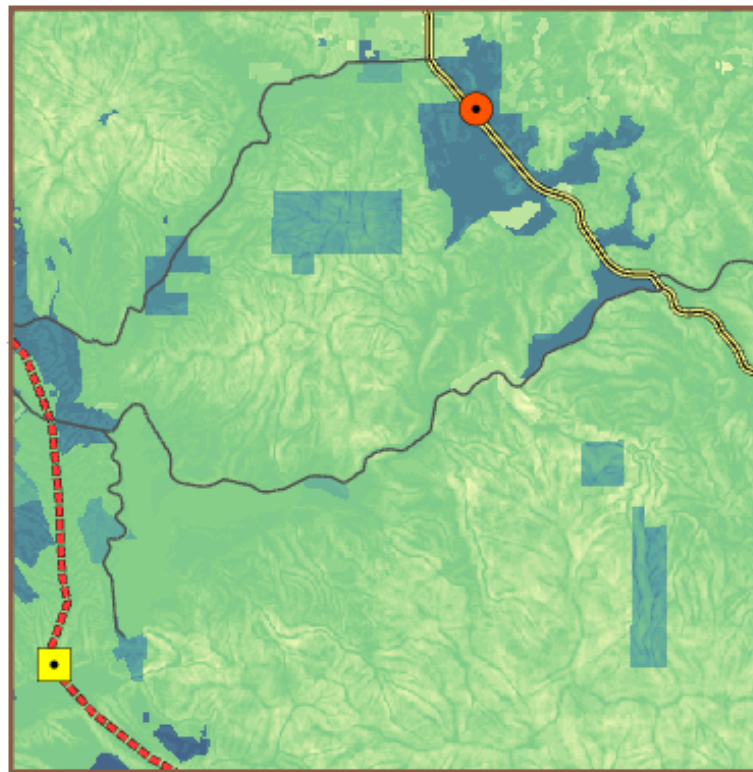


Figure 45. Cost surface displayed across the study area.

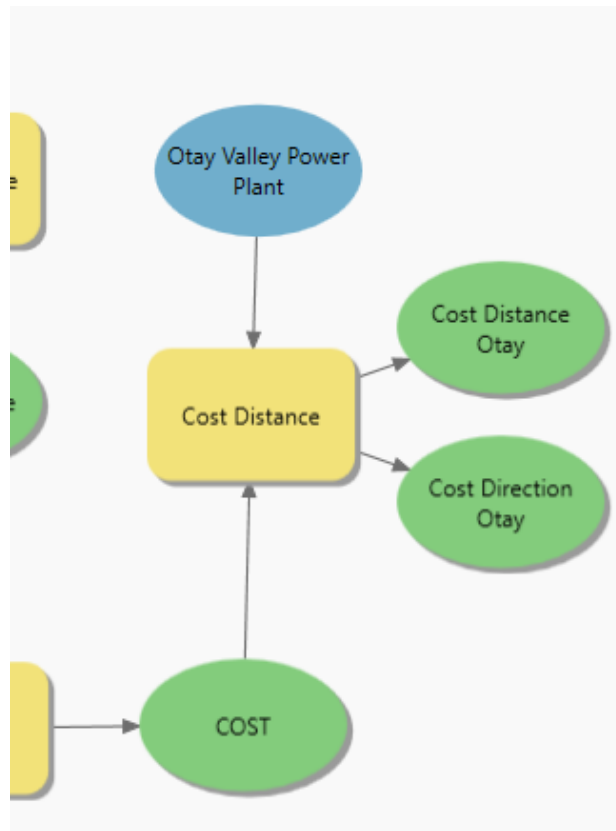


Figure 46. Model: Otag Valley Power Plant source into Cost Distance producing cost-distance, cost-direction and COST outputs.

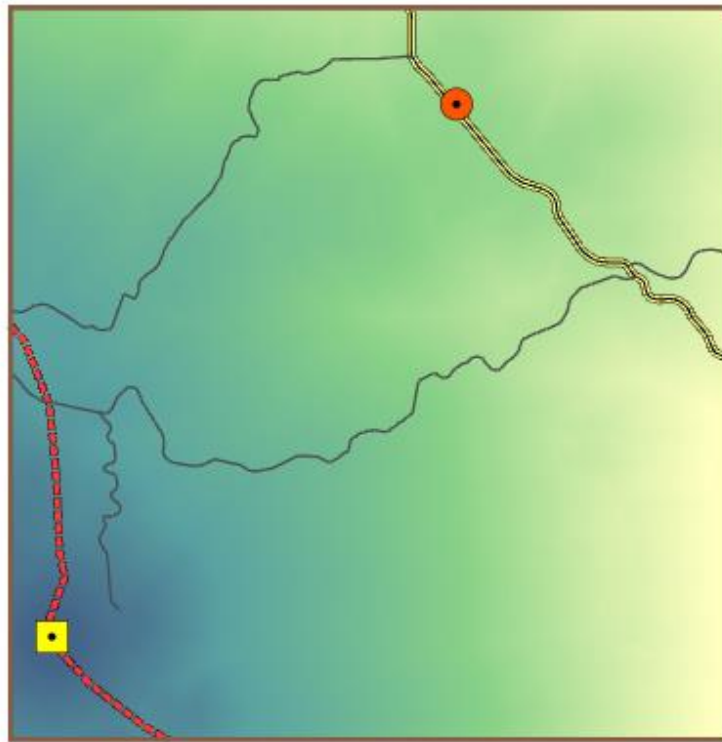


Figure 47. *Cost-weighted distance surface: accumulated cost of travel away from the power plant.*

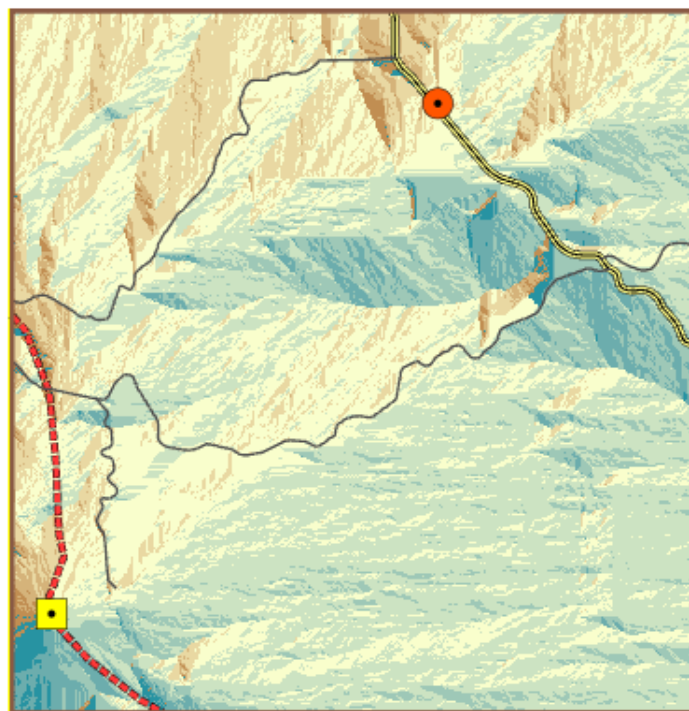


Figure 48. Cost-distance surface draped over a hillshade of the terrain.

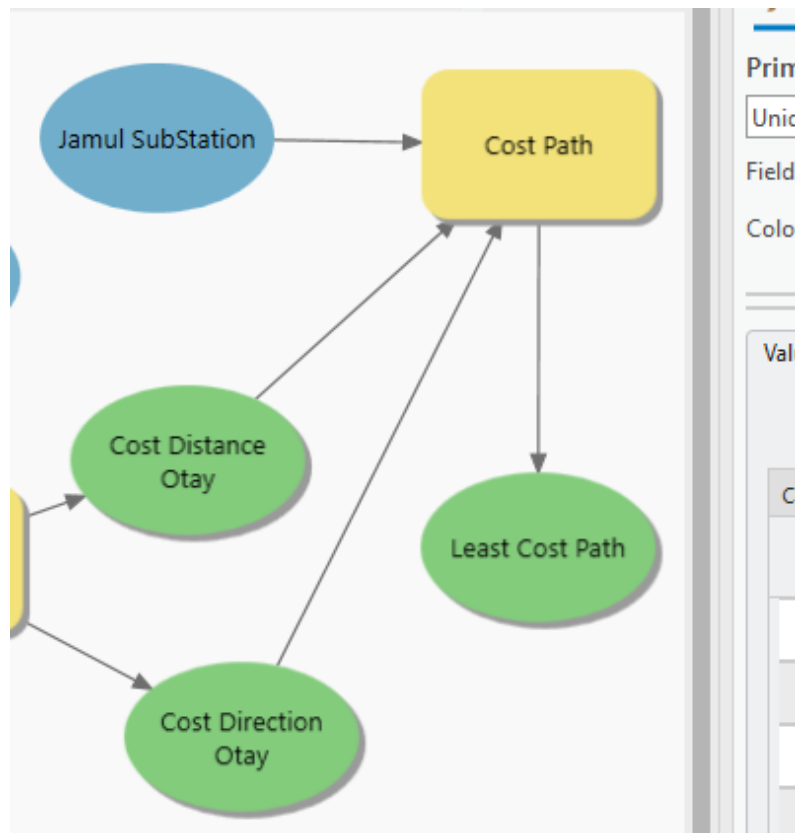


Figure 49. Model: Jamul Substation into Cost Path producing the Least Cost Path.

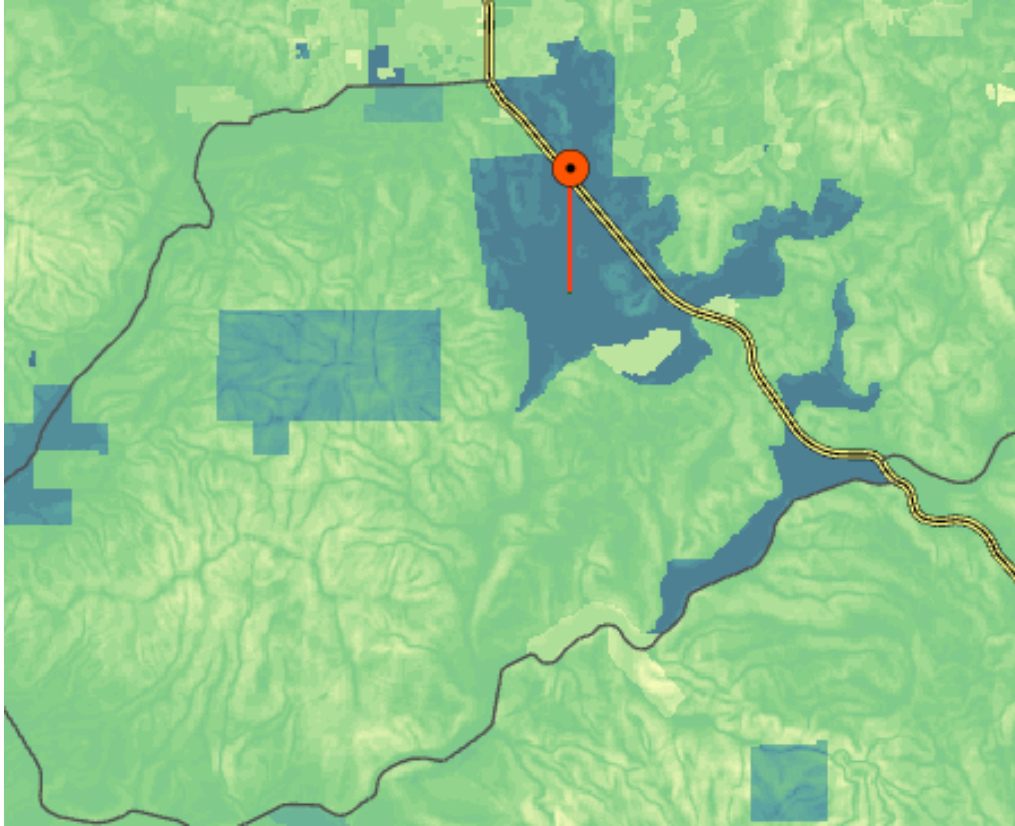


Figure 50. *Least-cost path result: the cheapest power-line route from the plant to the substation.*

4.3 Task 3 — Density Surfaces (Simple and Kernel Methods)

The third task used the Density function to map the concentration of incidents across a street network. Both methods supported by the tool employ a circular search area to make the calculation. The simple method divides the number of features that fall within the search neighbourhood by the area of that neighbourhood and assigns the result to the cell. The kernel method instead draws a smoothly curved surface around each point that falls from one at the point to zero at the neighbourhood boundary, and sums these kernels where they overlap; it generally produces a smoother, more interpretable surface.

Applying the kernel method to the burglary points produced a density surface in which the darker areas mark clear hotspots — clusters where incidents are most concentrated — while lighter areas indicate sparse activity.



Figure 51. *Street network of the study area used for the density analysis.*

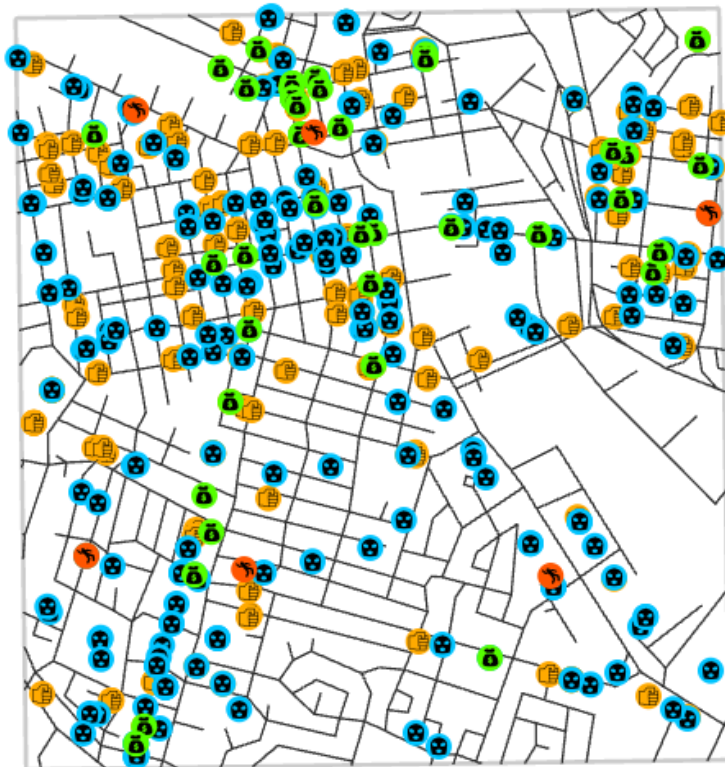


Figure 52. Incident point locations classified by type across the street network.



Figure 53. Burglary point locations used to build the density surface.

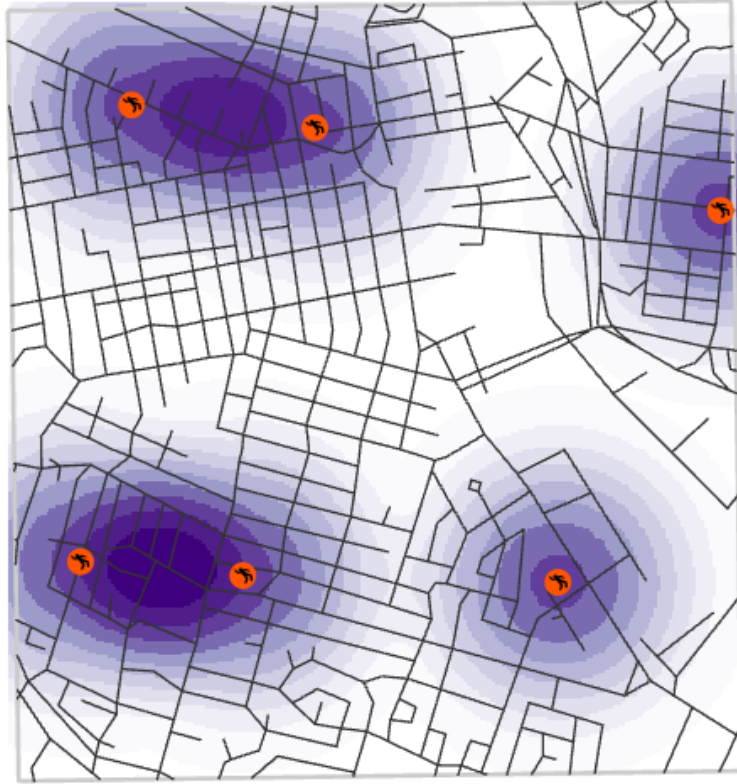


Figure 54. *Kernel density surface highlighting burglary hotspots.*



Figure 55. *Density surface shown with an adjusted classification scheme.*

4.4 Task 4 — Estimating Density Using Attributes

The final task refined the density analysis by weighting it with a feature attribute rather than treating every point equally. When a population or count field is supplied, the kernel function runs from the attribute value at the location of the point down to zero at the neighbourhood boundary, and the cell values become the sum of the attribute values in the overlapping kernels. Here the incident SCORE attribute was used, so locations carrying higher recorded values contributed proportionally more to the surface.

The resulting attribute-weighted surface gives a more realistic picture of intensity than the unweighted density, because it reflects not only where incidents occur but how significant each one is.



Figure 56. *Street-network base map for the attribute-weighted density analysis.*



Figure 57. Incident points symbolised by an attribute value.

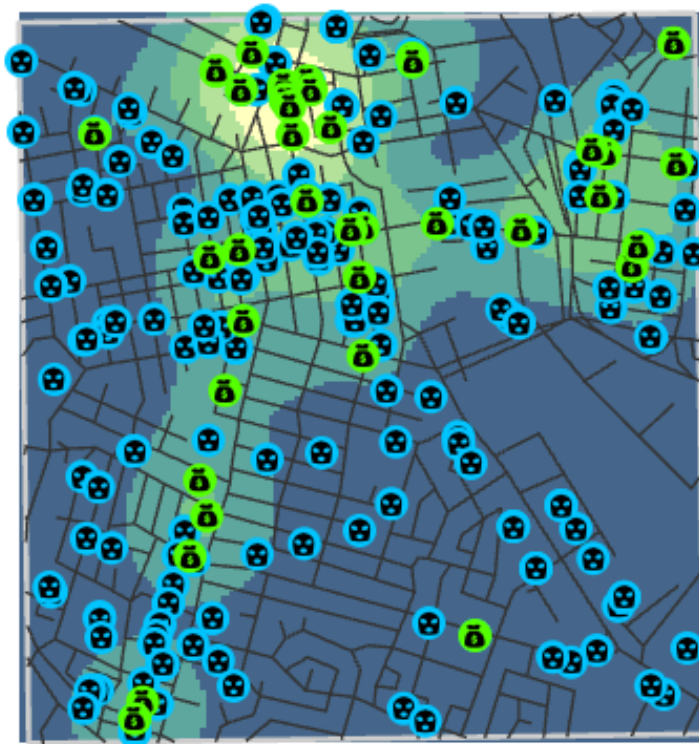


Figure 58. Attribute-weighted density surface with the incident points overlaid.

1:32,343 | 121.6534959°W 36.6624119°N

Grand theft

Field: Selection:

	OBJECTID *	Shape *	SCORE	OCCDATE1	OCCTIME1	RDATE	RTIME	ADDRESS	PARCEL
1	1	Point	100	19960621	00:39	19960621	00:39	W Lake St / Martella St	
2	2	Point	100	19960624	13:03	19960624	13:03	California St / E Lake St	
3	3	Point	100	19960627	17:00	19960627	17:18	E Alisal St / Prader St	
4	4	Point	100	19960628	13:00	19960628	14:07	29 California St	00220109
5	5	Point	100	19960630	21:00	19960705	16:47	E Market St / N Madeir...	
6	6	Point	100	19960709	20:50	19960709	20:54	19 E Romie Ln	00260120
7	7	Point	100	19960711	18:30	19960711	18:52	San Benito St / Smith St	

Figure 59. Grand-theft attribute table; the SCORE field is used to weight the density calculation.

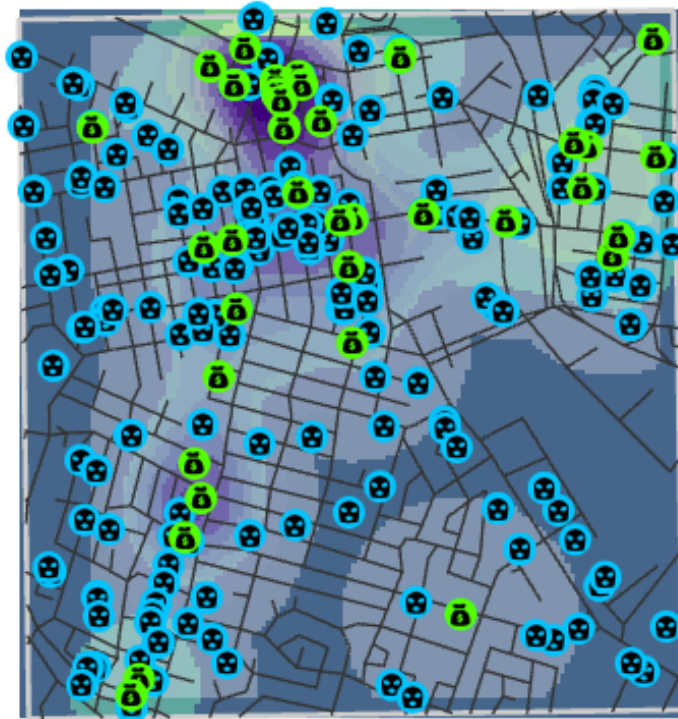


Figure 60. Attribute-weighted density surface across the study area.



Figure 61. Incident point distribution used in the weighted analysis.

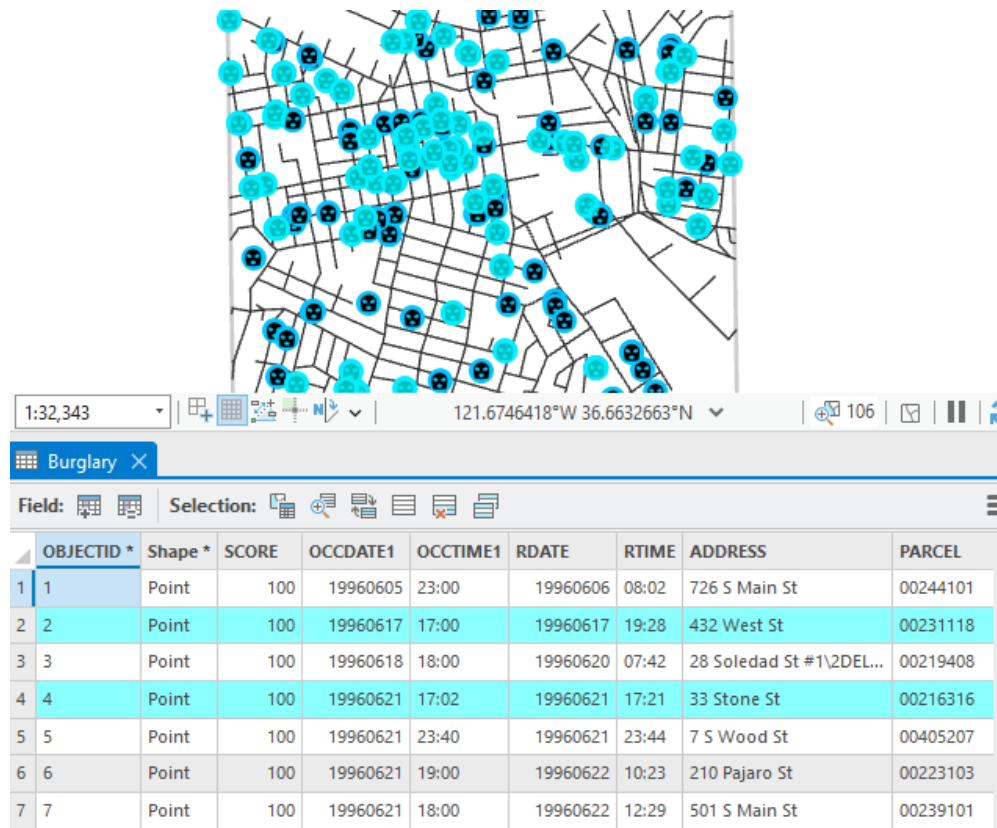


Figure 62. *Burglary attribute table feeding the weighted density calculation.*



Figure 63. *Attribute-weighted kernel density surface showing a concentrated hotspot.*

5. Discussion

The exercise demonstrated the practical difference between the two families of distance surface. Straight-line distance is simple and is often used directly in analysis, as in the air-ambulance example where travel is unobstructed. Cost-weighted distance is more realistic for movement over land, but its values are accumulated costs rather than true distances, so it is normally used indirectly as the input to a least-cost path analysis rather than read as a distance map. Building the cost surface also highlighted that the most demanding part of a cost analysis is deciding which factors matter and how to rank them, since a single final cost surface is required by the tool.

The density tasks showed how point observations can be turned into continuous surfaces that expose spatial patterns. The kernel method produced smoother, more readable hotspots than the simple method, and weighting the calculation by an attribute produced a surface that reflects the importance of each observation rather than a simple count.

6. Conclusion

All four tasks of Module 5 were completed successfully. Straight-line distance, direction and allocation surfaces were produced for the air-ambulance scenario; cost-weighted distance and direction surfaces were combined with a least-cost path analysis to route a power line between two facilities; and density surfaces were generated using the simple, kernel and attribute-weighted methods. The exercise provided a solid

working understanding of how ArcGIS Spatial Analyst represents proximity and concentration, and of how ModelBuilder can be used to assemble these steps into repeatable analytical workflows.